SURVIVAL ANALYSIS REPORT

Prognostic Factors in Advanced Lung Cancer: A Survival Analysis

Kaplan–Meier, Cox proportional hazards and parametric (Weibull) modelling of the NCCTG lung dataset in R

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Software: R with the `survival` package

Dataset: `survival::lung` (228 patients, 10 variables)

Bottom line. Female sex is associated with a roughly 45% lower hazard of death (HR 0.55, 95% CI 0.39 to 0.78), while each one-point worsening of physician-rated performance status (ph.ecog) is associated with a roughly 67% higher hazard (HR 1.67, 95% CI 1.31 to 2.14). Age and weight loss were not significant once these were accounted for. The Cox model assumptions held (global $p = 0.31$) but discrimination is modest (C-index 0.647). A Weibull model fitted best among the parametric options and agrees closely with the Cox model.

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1. Executive summary

This project studies how long patients with advanced lung cancer survive and which patient characteristics are linked to shorter or longer survival. Survival analysis is the right family of methods because the outcome is a time to an event (death) and many patients are still alive at the end of follow-up (censored), which ordinary regression cannot handle correctly.

Key findings

- **Data:** 228 patients; 165 died (72.4%) and 63 were censored. After removing rows with missing predictors, 213 patients and 151 deaths were used for regression models.
- **Kaplan–Meier and log-rank:** median survival was 270 days for men and 426 days for women. The difference is statistically significant (log-rank $\chi^2 = 10.3$, 1 df, $p = 0.0013$).
- **Cox model:** female sex (HR 0.554, $p = 0.0008$) and ph.ecog (HR 1.674, $p < 0.0001$) were significant. Age (HR 1.013, $p = 0.165$) and weight loss (HR 0.991, $p = 0.176$) were not.
- **Proportional hazards assumption:** satisfied for every covariate and globally (global $p = 0.314$). Sex was the closest to the boundary ($p = 0.10$).
- **Predictive strength:** C-index 0.647 (SE 0.026), meaning modest ability to rank patients by risk.
- **Parametric models:** the Weibull model had the lowest AIC (2106.98 vs 2128.13 exponential and 2136.39 log-normal). Its estimated scale of 0.716 implies a hazard that increases over time, and its predictions track the Cox curve closely.
- **Practical message:** a patient's functional status (ph.ecog) and sex carry most of the prognostic information in this model; age and recent weight loss add little once those are known.

2. Background and objectives

The `lung` data come from a North Central Cancer Treatment Group (NCCTG) study of patients with advanced lung cancer and are distributed with the R `survival` package. Time is measured in days from enrolment until death or the end of observation.

Objectives

1. Describe the data and handle missing values.
2. Estimate and compare survival curves for men and women.
3. Quantify the independent effect of age, sex, performance status and weight loss on the risk of death.
4. Check whether the Cox model's assumptions are valid and how well it discriminates.
5. Fit parametric survival models, choose the best one and compare its predictions with the Cox model.

3. Dataset and variable dictionary

The dataset has 228 rows and 10 variables. The table lists each variable, its meaning and coding, how many values are missing, and the role it plays in this analysis.

Variable	Meaning and coding	Missing	Role in this project
`inst`	Institution (hospital) code	1	Not used
`time`	Survival time in days from enrolment to death or censoring. Observed range 5 to 1022 days; median 255.5	0	Outcome (time part)
`status`	Event indicator as supplied: 1 = censored, 2 = dead (63 censored, 165 dead)	0	Recoded to `event` (1 = death, 0 = censored)
`age`	Age in years. Range 39 to 82; mean 62.45; median 63	0	Covariate (continuous)
`sex`	Supplied as 1 = male, 2 = female (138 male, 90 female); recoded to a labelled factor with Male as reference	0	Grouping variable for KM and log-rank; covariate in models
`ph.ecog`	ECOG performance score rated by the physician: 0 = asymptomatic, 1 = symptomatic but completely ambulatory, 2 = in bed less than 50% of the day, 3 = in bed more than 50% of the day but not bedbound (4 = bedbound is possible on the scale). Higher = worse. Median 1	1	Covariate (treated as numeric, one-point trend)
`ph.karno`	Karnofsky performance score rated by the physician (0 to 100, higher = better). Mean 81.9	1	Not used
`pat.karno`	Karnofsky performance score rated by the patient (0 to 100, higher = better). Mean 80.0	3	Not used
`meal.cal`	Calories consumed at meals. Median 975	47	Not used (too many missing values)
`wt.loss`	Weight loss in the last six months (pounds). Range -24 to 68; median 7; negative = gain	14	Covariate (continuous)
`event` (derived)	1 if status = 2 else 0	0	Outcome (event part)

Why these four covariates?

Age, sex, performance status and weight loss are the standard, clinically meaningful prognostic factors available with few missing values. The other performance scores (ph.karno, pat.karno) measure essentially the same construct as ph.ecog, so including all of them would add collinearity without adding information, and meal.cal was missing for 47 patients (20.6%), which would have discarded too much data.

4. Methods: what was done, why, and what it tells us

Step	Method / R function	Why it was done	What it tells us
1. Inspect data	`str`, `summary`, `head`, `dim`	Confirm structure, ranges and coding before modelling	Variable types, plausible ranges, missingness
2. Missing values	`colSums(is.na())`, `table(status)`	Regression functions drop incomplete rows silently; the analyst should know how many and which	meal.cal (47) and wt.loss (14) have the most missing values; 165 deaths vs 63 censored
3. Recode	`event`, factor `sex`	Models need 0/1 event coding and readable labels	Correct reference group (Male) and event definition
4. Complete-case sets	`complete.cases`, `na.omit`	KM needs only time, event, sex; the models need all covariates	228 patients for KM; 213 patients with 151 deaths for models
5. Kaplan–Meier	`survfit(Surv(time, event) ~ sex)`	Non-parametric estimate of the survival function that correctly uses censored patients	Survival probability over time and median survival per group
6. Log-rank test	`survdiff`	Tests whether two survival curves are statistically different	Evidence that survival differs between men and women
7. Cox model	`coxph`	Estimates the effect of several covariates at once without assuming a baseline distribution	Adjusted hazard ratios with confidence intervals
8. Proportional hazards check	`cox.zph`, Schoenfeld residual plots	The Cox model assumes each hazard ratio is constant over time; if not, estimates are misleading	Whether the model is valid; whether effects change with time
9. C-index	`summary(cox)$concordance`	Significant coefficients do not guarantee good prediction; discrimination must be measured	Probability that the model ranks a pair of patients correctly (0.5 = chance, 1 = perfect)
10. Likelihood-ratio tests	`drop1(..., test = "Chisq")`	Tests each variable by removing it from the model; more reliable than Wald z-tests in small samples	Which variables genuinely improve the fit (also gives AIC without each term)
11. Parametric models	`survreg` with Weibull, exponential, log-normal	Give a full survival distribution, smooth predictions and extrapolation beyond observed data	Time ratios and the shape of the hazard over time
12. AIC comparison	`AIC()`	Chooses among non-nested parametric models, balancing fit and complexity	Weibull fits best
13. Prediction	`survfit(cox, newdata)` and Weibull formula	Shows how the two modelling approaches agree for a concrete patient	Predicted survival curves for a 65-year-old woman with ph.ecog 1

5. Data preparation results

Item	Result
Patients in original data	228
Deaths (status = 2)	165 (72.4%)
Censored (status = 1)	63 (27.6%)
Patients in Kaplan–Meier analysis	228 (no missing values in time, event or sex)
Patients in Cox and parametric models	213 (15 patients, 6.6%, removed for missing age, sex, ph.ecog or wt.loss)
Deaths in model dataset	151

Note on missing data. A complete-case analysis assumes the removed patients are not systematically different from those retained. Only 6.6% were removed, so the risk of serious bias is small, but multiple imputation would be the more rigorous option.

6. Results and interpretation

6.1 Kaplan–Meier survival by sex

Why: to see how survival evolves over time in each sex without assuming any model, using all 228 patients and handling censored observations properly.

Group	n	Deaths	Median survival (days)	95% CI
Male	138	112	270	212 to 310
Female	90	53	426	348 to 550

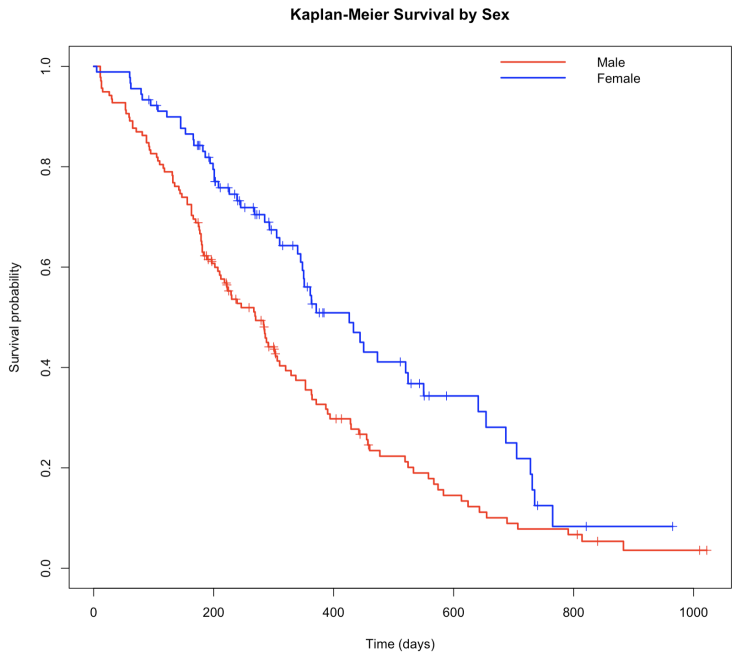


Figure 1. *Kaplan–Meier survival curves by sex (tick marks are censored patients).*

Interpretation. The female curve lies above the male curve for nearly the whole follow-up. Half of the men had died by about 270 days, whereas half of the women had died only by about 426 days, a difference of roughly 5 months. The confidence intervals for the two medians do not overlap, which already hints at a real difference. The curves come close together after about 750 days, when few patients remain at risk and estimates are imprecise.

6.2 Log-rank test

Why: the plot shows a gap, but the log-rank test checks whether that gap could be due to chance. It compares the observed number of deaths in each group with the number expected if both groups had identical survival.

Group	N	Observed	Expected	$(O-E)^2/E$	$(O-E)^2/V$
Male	138	112	91.6	4.55	10.3
Female	90	53	73.4	5.68	10.3

Chi-square $= 10.33$ on 1 degree of freedom, $\mathbf{p = 0.0013}$.

Interpretation. Men experienced 112 deaths when only 91.6 were expected under equal survival, while women experienced 53 deaths against 73.4 expected. With $p = 0.0013$ the null hypothesis of identical survival curves is rejected: survival differs significantly by sex. The log-rank test is unadjusted, so the Cox model is needed to see whether the difference persists after accounting for age, performance status and weight loss.

6.3 Cox proportional hazards model

Why: the Cox model estimates the effect of each covariate on the hazard of death while holding the others constant, and it does not require choosing a survival distribution. The hazard ratio (HR) is $\exp(\text{coefficient})$: an HR above 1 means higher risk of death, below 1 means lower risk.

Covariate	Coef	HR	95% CI for HR	z	p-value
age (per year)	0.0134	1.013	0.995 to 1.033	1.39	0.165
sex: Female vs Male	-0.5908	0.554	0.393 to 0.781	-3.37	0.0008 ***
ph.ecog (per point)	0.5151	1.674	1.308 to 2.143	4.09	< 0.0001 ***
wt.loss (per lb)	-0.0090	0.991	0.978 to 1.004	-1.35	0.176

$n = 213$, events $= 151$. Likelihood-ratio test $= 31.02$ on 4 df ($p = 3 \times 10^{-6}$); Wald test $= 29.94$ ($p = 5 \times 10^{-6}$); score (log-rank) test $= 30.65$ ($p = 4 \times 10^{-6}$). All three overall tests agree that the model as a whole is highly significant.

Interpretation of each covariate

- **Sex (HR 0.554).** At any moment, a woman has about 45% lower hazard of death than a man of the same age, performance status and weight loss. The interval (0.39 to 0.78) lies entirely below 1, so the effect is significant and consistent with the Kaplan–Meier result.
- **ph.ecog (HR 1.674).** Each one-point worsening of performance status raises the hazard of death by about 67% (interval 31% to 114%). This is the strongest prognostic factor. A patient with score 2 compared with score 0 has roughly $1.67^2 \approx 2.8$ times the hazard, assuming the effect is linear on the log-hazard scale.
- **Age (HR 1.013).** Each extra year of age is associated with about a 1.3% higher hazard, but the interval (0.995 to 1.033) includes 1 and $p = 0.165$, so age is not a significant independent predictor in this sample.
- **Weight loss (HR 0.991).** The estimate is slightly below 1 (unexpected clinically) but is statistically indistinguishable from no effect ($p = 0.176$). It should not be read as protective.

6.4 Proportional hazards assumption

Why: the Cox model assumes each HR stays constant over time. If, for example, the female advantage shrank at later times, a single HR would be a misleading average. The test examines whether scaled Schoenfeld residuals show a trend against time; the plots show the estimated coefficient $\beta(t)$ with a smooth line and confidence band. If a horizontal line at the estimated coefficient fits inside the confidence band, the effect is consistent with being constant over time.

Term	Chi-square	df	p-value	Conclusion
age	0.4353	1	0.51	Assumption holds
sex	2.6731	1	0.10	Holds, borderline
ph.ecog	1.6355	1	0.20	Assumption holds
wt.loss	0.0457	1	0.83	Assumption holds
GLOBAL	4.7516	4	0.31	Assumption holds overall

No p-value is below 0.05, so there is no statistical evidence that the hazard ratios change over time, and the global test ($p = 0.314$) supports using a standard Cox model.

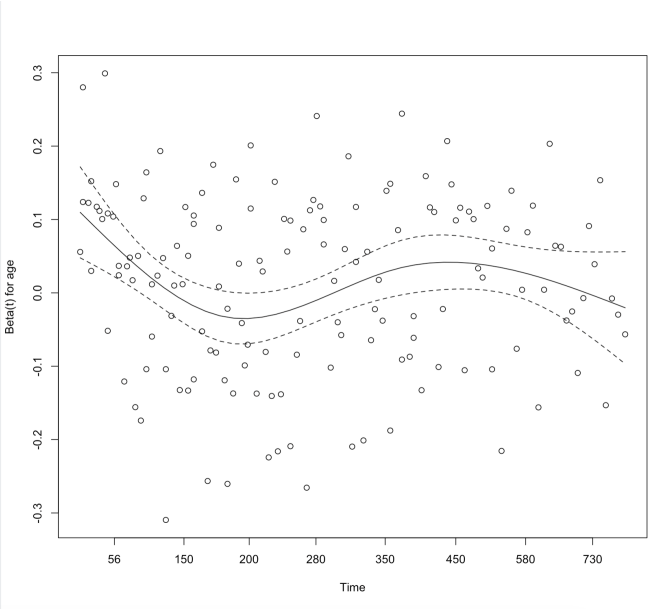


Figure 2. *Scaled Schoenfeld residuals for age.*

Age. The smooth line starts higher (about 0.1 at the earliest times) and then settles near zero, but a horizontal line at the estimated coefficient (0.013) fits inside the confidence band everywhere except the very earliest times; the test $p = 0.51$ confirms no meaningful time trend.

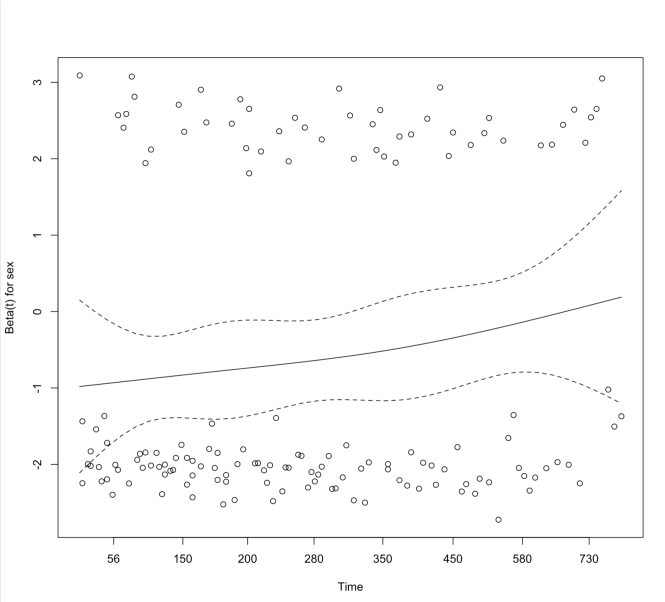


Figure 3. *Scaled Schoenfeld residuals for sex.*

Sex. Residuals of a binary variable split into two bands, which is normal. The smooth line rises gently from about -1.0 toward 0.2, suggesting the female advantage might weaken at late times, but the band is wide, a horizontal line at the estimated coefficient (-0.59) stays inside it throughout, and $p = 0.10$ is not significant. This is the covariate to watch if the dataset were larger.

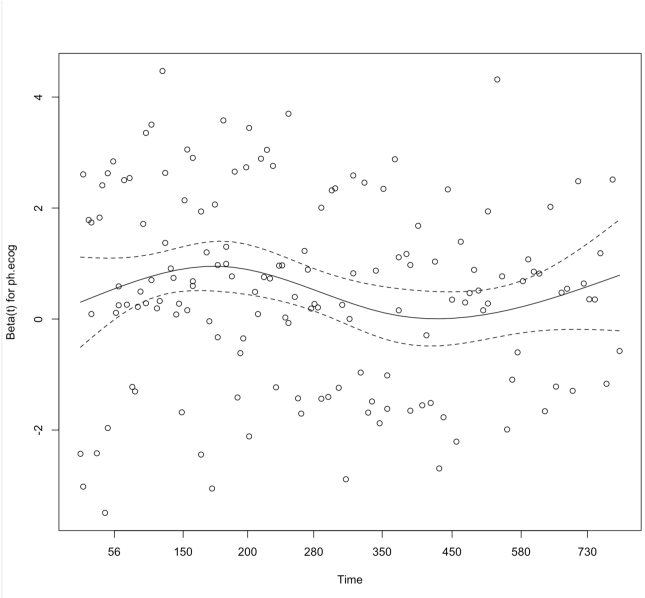


Figure 4. *Scaled Schoenfeld residuals for ph.ecog.*

ph.ecog. The line undulates between roughly 0 and 1 around the estimated coefficient (0.515), but a horizontal line at that value stays essentially within the band; $p = 0.20$ indicates no significant change in the effect of performance status over time.

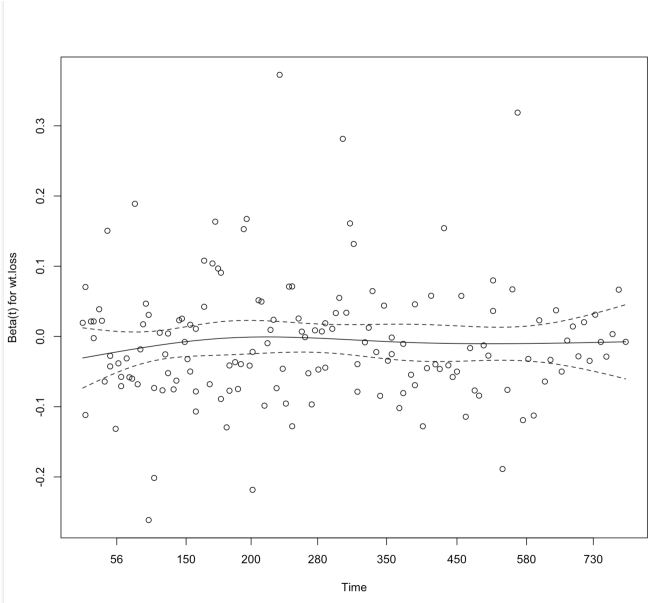


Figure 5. *Scaled Schoenfeld residuals for wt.loss.*

Weight loss. The line is essentially flat and close to the estimated coefficient (-0.009), in line with the large p-value of 0.83.

6.5 Discrimination: concordance index

Why: a variable can be statistically significant yet the model can still be poor at telling patients apart. The concordance index is the proportion of comparable patient pairs in which the patient with the higher predicted risk actually dies first.

Result: C-index $= \mathbf{0.647}$ (SE 0.026), giving an approximate 95% interval of 0.60 to 0.70.

Interpretation. The model ranks patients correctly about 65% of the time, better than a coin toss (0.5) but well below strong clinical models (usually 0.7 to 0.8). With only four covariates this is expected: survival in advanced lung cancer depends on many factors not recorded here (tumour stage, treatment, laboratory values). The model is useful for explaining associations and giving rough risk groups, but not for precise individual prognosis. The C-index was computed on the same data used to fit the model, so it is likely slightly optimistic; cross-validation or bootstrapping would give a corrected value.

6.6 Likelihood-ratio tests for each predictor (drop1)

Why: Wald z-tests are approximations. `drop1` refits the model without each variable in turn and tests the loss of fit with a likelihood-ratio (LR) chi-square. The AIC column shows the model AIC after removing that term; a value higher than the full model (1327.0) means the term was useful.

Term removed	Df	AIC after removal	LRT	p-value
none (full model)		1327.0		
age	1	1327.0	1.966	0.161
sex	1	1337.0	11.988	0.0005 ***
ph.ecog	1	1341.6	16.603	< 0.0001 ***
wt.loss	1	1326.9	1.892	0.169

Interpretation. Removing sex or ph.ecog worsens AIC by about 10 and 15 points, so both are essential. Removing age leaves AIC unchanged and removing wt.loss slightly improves it (1326.9), so neither adds explanatory value. A simpler model with only sex and ph.ecog would be nearly as good and easier to interpret. These results agree with the Wald tests in the Cox summary.

6.7 Parametric survival models (accelerated failure time)

Why: the Cox model leaves the baseline hazard unspecified, so it cannot give a smooth survival curve or extrapolate beyond the last observed event. Parametric models assume a distribution for survival time, which yields full survival curves, interpretable time ratios, and the shape of the hazard. Three distributions were compared: exponential (constant hazard), Weibull (monotonically increasing or decreasing hazard) and log-normal.

Model comparison by AIC

Model	Parameters (df)	AIC	Difference from best
Exponential	5	2128.13	$+21.15$
Weibull	6	**2106.98**	0 (best)
Log-normal	6	2136.39	$+29.41$

A lower AIC is better, and differences above 10 are generally considered strong evidence. The Weibull model is clearly preferred. **Caution:** these AIC values (about 2100) cannot be compared with the Cox AIC (about 1327), because the Cox model uses a partial likelihood and the parametric models use a full likelihood.

Weibull model results

Term	Coef	Std. error	z	p-value	Time ratio
(Intercept)	6.7574	0.4351	15.53	$< 2 \times 10^{-16}$	–
age	-0.0089	0.0069	-1.29	0.196	0.991
sex: Female	0.4162	0.1270	3.28	0.001	1.516
ph.ecog	-0.3664	0.0904	-4.05	5.0×10^{-5}	0.693
wt.loss	0.0061	0.0047	1.30	0.194	1.006
Log(scale)	-0.3338	0.0638	-5.24	1.6×10^{-7}	–

Scale $= 0.716$; log-likelihood of the model $= -1047.5$ versus -1062.6 for the intercept-only model; chi-square $= 30.14$ on 4 df, $p = 4.6 \times 10^{-6}$; $n = 213$.

How to read the Weibull output

- In an accelerated failure time (AFT) model, coefficients act on survival **time**, not on the hazard. A positive coefficient lengthens survival; a negative one shortens it. This is the opposite sign to the Cox coefficients.
- **Sex:** $\exp(0.416) = 1.52$, so, other things equal, women survive about 52% longer than men.
- **ph.ecog:** $\exp(-0.366) = 0.69$, so each additional point of performance score shortens survival time by about 31%.
- **Age and wt.loss** are again not significant ($p = 0.196$ and 0.194).
- **Shape:** the shape parameter is $1/\text{scale} = 1/0.716 = 1.40$. A shape above 1 means the hazard of death rises over time, and the highly significant Log(scale) ($z = -5.24$) shows the scale differs from 1, which is why the exponential model, which forces constant hazard, fits worse.

Agreement with the Cox model

A Weibull model can also be expressed as a proportional hazards model with HR $= \exp(-\text{coef}/\text{scale})$. This gives HR $= \exp(-0.416/0.716) = 0.56$ for female sex and HR $= \exp(0.366/0.716) = 1.67$ for ph.ecog, almost identical to the Cox estimates of 0.554 and 1.674. The two independent approaches therefore tell the same story, which strengthens confidence in the findings.

6.8 Predicted survival for an example patient

Why: to show what the models mean in practice. The example patient is a 65-year-old woman with ph.ecog $= 1$ and median weight loss. The Cox curve is a step function derived from the data; the Weibull curve is the smooth parametric prediction

$$S(t) = \exp\left[-\left(\frac{t}{\exp(\mathsf{lp})}\right)^{1/\text{scale}}\right].$$

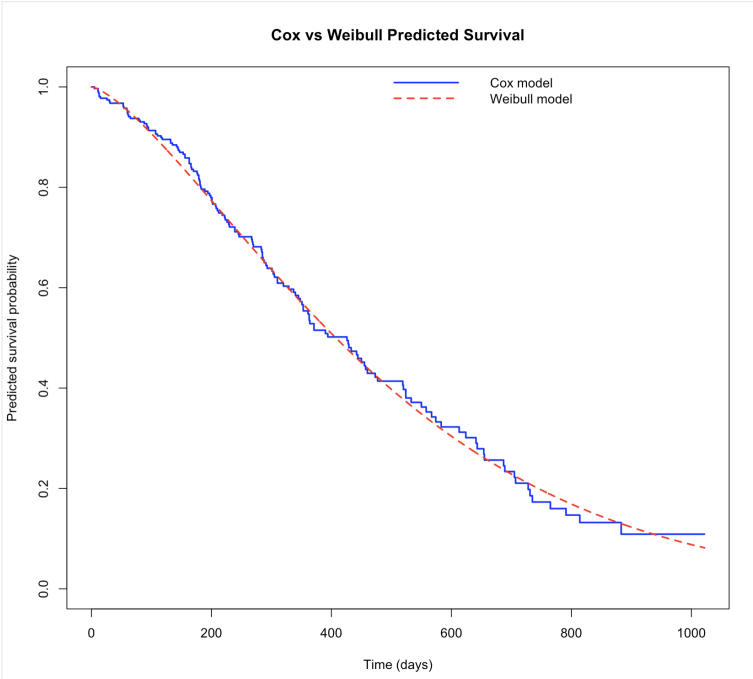


Figure 6. *Predicted survival for the example patient: Cox (solid blue) vs Weibull (dashed red).*

Interpretation. The two curves almost overlap from 0 to about 700 days, which confirms the Weibull model describes the data well. Predicted survival falls to 50% at roughly 450 days (read from the plot). Beyond about 700 days the Cox curve is flat at about 0.11 after the last observed death, because a non-parametric estimate cannot move without events, whereas the Weibull curve keeps declining smoothly toward 0.08 at 1022 days. Beyond the observed data, the Weibull prediction is the more plausible, but it rests on the assumed distribution.

7. Conclusions

1. Survival differs by sex: median survival is 426 days for women versus 270 days for men (log-rank $p = 0.0013$).
2. After adjusting for age, performance status and weight loss, female sex remains independently protective (HR 0.554, 95% CI 0.393 to 0.781).
3. Worse physician-rated performance status is the strongest adverse factor (HR 1.674 per point, 95% CI 1.308 to 2.143).
4. Age and recent weight loss are not significant in this model and could be dropped without loss of fit (drop1 results).
5. The proportional hazards assumption is satisfied (global $p = 0.314$), so the hazard ratios can be interpreted as constant over follow-up.
6. The model discriminates modestly (C-index 0.647), so it explains group-level risk better than it predicts individual outcomes.
7. The Weibull model is the best parametric choice (AIC 2106.98) with an increasing hazard over time, and it reproduces the Cox results (HR 0.56 and 1.67), giving consistent evidence from two modelling frameworks.

8. Limitations and recommendations

Limitations

- Complete-case analysis removed 15 patients (6.6%); meal.cal, ph.karno and pat.karno were not used. Multiple imputation would be more rigorous.
- The C-index was not corrected for optimism, and no external validation was done.
- Observational registry-type data cannot prove causation; unmeasured factors (stage, treatment, comorbidity) may confound the associations. The sex difference may partly reflect such factors.

Recommendations for further work

- Refit a reduced model with sex and ph.ecog only and compare AIC and C-index.
- Check functional form with martingale residuals or splines, and treat ph.ecog as a factor.
- Use multiple imputation instead of complete-case analysis.
- Validate with bootstrap or cross-validated C-index, and add a time-dependent AUC or calibration plot.
- Consider stratifying or adding a time interaction for sex if the borderline non-proportionality persists in larger data.

9. Glossary

Term	Meaning
Survival time	Time from the start of observation (enrolment) until the event or the end of observation
Censoring	The patient was still alive, or lost to follow-up, so the exact death time is unknown but is known to exceed the observed time
Survival function $S(t)$	Probability of surviving beyond time t
Hazard	Instantaneous risk of dying at time t among those still alive
Hazard ratio (HR)	Ratio of hazards between two groups; 1 = no effect, above 1 = higher risk, below 1 = lower risk
Kaplan–Meier estimator	Non-parametric estimate of $S(t)$ that accounts for censoring
Log-rank test	Test that two or more survival curves are identical
Cox model	Semi-parametric regression for the hazard that makes no assumption about the baseline hazard
Proportional hazards	Assumption that hazard ratios do not change over time
Schoenfeld residuals	Residuals used to test and visualise the proportional hazards assumption
C-index (concordance)	Probability that the patient predicted to be at higher risk dies first; 0.5 = chance, 1 = perfect
AIC	Akaike information criterion; lower values indicate a better balance of fit and simplicity

Term	Meaning
AFT model	Accelerated failure time model; covariates multiply survival time by a constant (time ratio)
Weibull distribution	Flexible survival distribution; shape above 1 = hazard rising over time, below 1 = falling, equal to 1 = exponential
ECOG score	Physician-rated scale of functional status, 0 (fully active) to 4 (bedbound)

A. Complete R code

The script below reproduces every result and figure in this report. Run it from top to bottom in R after installing the `survival` package.

```
# ============================================================
# Survival analysis of the NCCTG advanced lung cancer data
# ============================================================
library(survival)

# 1. Load and inspect the dataset
lung <- survival::lung
str(lung)
summary(lung)
dim(lung)
head(lung)
cat("Missing values in original data:\n")
print(colSums(is.na(lung)))
cat("Status counts:\n")
print(table(lung$status))

# 2. Prepare the data
# status = 2 means death; status = 1 means censored
lung$event <- as.integer(lung$status == 2)
# Original coding: 1 = Male, 2 = Female
lung$sex <- factor(lung$sex, levels = c(1, 2),
                   labels = c("Male", "Female"))

# Data for Kaplan-Meier analysis
km_data <- lung[complete.cases(lung[, c("time", "event", "sex")]), ]
# Complete cases for regression and parametric models
d <- na.omit(lung[, c("time", "event", "age", "sex", "ph.ecog", "wt.loss")])
cat("Observations for KM analysis:", nrow(km_data), "\n")
cat("Complete observations for regression:", nrow(d), "\n")
cat("Deaths in regression dataset:", sum(d$event), "\n")
stopifnot(nrow(km_data) > 0, nrow(d) > 0)

# 3. Kaplan-Meier curves by sex
km <- survfit(Surv(time, event) ~ sex, data = km_data)
print(km)
plot(km, col = c("red", "blue"), lwd = 2,
     xlab = "Time (days)", ylab = "Survival probability",
     main = "Kaplan-Meier Survival by Sex",
     mark.time = TRUE, conf.int = FALSE)
legend("topright", legend = c("Male", "Female"),
       col = c("red", "blue"), lwd = 2, bty = "n")

# 4. Log-rank test
lr <- survdiff(Surv(time, event) ~ sex, data = km_data)
p_logrank <- pchisq(lr$chisq, df = length(lr$n) - 1, lower.tail = FALSE)
print(lr)
cat("Log-rank chi-square:", lr$chisq, "\n")
cat("Log-rank p-value:", p_logrank, "\n")

# 5. Cox proportional hazards model
cox_model <- coxph(Surv(time, event) ~ age + sex + ph.ecog + wt.loss,
                   data = d, x = TRUE)
print(summary(cox_model))

# Hazard ratios and 95% confidence intervals
cox_results <- exp(cbind(HR = coef(cox_model), confint(cox_model)))
cat("Hazard ratios and 95% confidence intervals:\n")
print(cox_results)

# 6. Proportional hazards assumption (Schoenfeld residuals)
ph_test <- cox.zph(cox_model)
print(ph_test)
plot(ph_test)          # one plot per covariate

# 7. Model discrimination: concordance index
c_index <- summary(cox_model)$concordance
cat("C-index:", c_index[1], "\n")
```

```r
cat("Standard error:", c_index[2], "\n")

# 8. Likelihood-ratio test for each predictor
factor_test <- drop1(cox_model, test = "Chisq")
cat("Likelihood-ratio tests for individual predictors:\n")
print(factor_test)

# 9. Parametric (accelerated failure time) models
weibull_model <- survreg(Surv(time, event) ~ age + sex + ph.ecog + wt.loss,
                         data = d, dist = "weibull")
cat("Weibull model results:\n")
print(summary(weibull_model))

exponential_model <- survreg(Surv(time, event) ~ age + sex + ph.ecog + wt.loss,
                             data = d, dist = "exponential")
lognormal_model <- survreg(Surv(time, event) ~ age + sex + ph.ecog + wt.loss,
                           data = d, dist = "lognormal")

# 10. Compare parametric models using AIC
aic_results <- AIC(exponential_model, weibull_model, lognormal_model)
cat("Parametric model comparison using AIC:\n")
print(aic_results)

# 11. Predicted survival for an example patient
patient <- data.frame(
  age = 65,
  sex = factor("Female", levels = levels(d$sex)),
  ph.ecog = 1,
  wt.loss = median(d$wt.loss)
)

# Cox prediction
cox_curve <- survfit(cox_model, newdata = patient)

# Weibull AFT prediction
times <- seq(0, max(d$time), by = 1)
lp <- as.numeric(predict(weibull_model, newdata = patient, type = "lp"))
scale_param <- weibull_model$scale
weibull_survival <- exp(-(times / exp(lp))^(1 / scale_param))

# Plot both predicted curves
plot(cox_curve, col = "blue", lwd = 2, conf.int = FALSE,
     xlab = "Time (days)", ylab = "Predicted survival probability",
     main = "Cox vs Weibull Predicted Survival",
     xlim = c(0, max(d$time)), ylim = c(0, 1))
lines(times, weibull_survival, col = "red", lwd = 2, lty = 2)
legend("topright", legend = c("Cox model", "Weibull model"),
       col = c("blue", "red"), lwd = 2, lty = c(1, 2), bty = "n")

# 12. Project summary
cat("\n========== PROJECT SUMMARY ==========\n")
cat("Log-rank p-value:", signif(p_logrank, 4), "\n")
cat("C-index:", round(c_index[1], 3), "\n")
cat("Cox model observations:", nrow(d), "\n")
cat("Cox model deaths:", sum(d$event), "\n")
cat("Global PH test p-value:", signif(ph_test$table["GLOBAL", "p"], 4), "\n")
cat("\nAIC comparison:\n")
print(aic_results)
cat("\nScript completed.\n")
```

B. Key R console output

B.1 Kaplan–Meier and log-rank

```
Call: survfit(formula = Surv(time, event) ~ sex, data = km_data)

             n events median 0.95LCL 0.95UCL
sex=Male   138    112    270     212     310
sex=Female  90     53    426     348     550

Call:
survdiff(formula = Surv(time, event) ~ sex, data = km_data)

             N Observed Expected (O-E)^2/E (O-E)^2/V
sex=Male   138      112     91.6      4.55      10.3
sex=Female  90       53     73.4      5.68      10.3

 Chisq= 10.3  on 1 degrees of freedom, p= 0.001
```

B.2 Cox model summary

```
  n= 213, number of events= 151

               coef exp(coef)  se(coef)      z Pr(>|z|)
age        0.013369  1.013459  0.009628  1.389 0.164951
sexFemale -0.590775  0.553898  0.175339 -3.369 0.000754 ***
ph.ecog    0.515111  1.673824  0.125988  4.089 4.34e-05 ***
wt.loss   -0.009006  0.991034  0.006658 -1.353 0.176135

          exp(coef) exp(-coef) lower .95 upper .95
age          1.0135     0.9867    0.9945    1.0328
sexFemale    0.5539     1.8054    0.3928    0.7811
ph.ecog      1.6738     0.5974    1.3076    2.1427
wt.loss      0.9910     1.0090    0.9782    1.0041

Concordance= 0.647  (se = 0.026 )
Likelihood ratio test= 31.02  on 4 df,   p=3e-06
Wald test            = 29.94  on 4 df,   p=5e-06
Score (logrank) test = 30.65  on 4 df,   p=4e-06
```

B.3 Proportional hazards test (cox.zph)

```
         chisq df    p
age     0.4353  1 0.51
sex     2.6731  1 0.10
ph.ecog 1.6355  1 0.20
wt.loss 0.0457  1 0.83
GLOBAL  4.7516  4 0.31
```

B.4 Likelihood-ratio tests (drop1)

```
        Df    AIC     LRT  Pr(>Chi)
<none>     1327.0
age      1 1327.0  1.9661 0.1608636
sex      1 1337.0 11.9881 0.0005354 ***
ph.ecog  1 1341.6 16.6029 4.608e-05 ***
wt.loss  1 1326.9  1.8915 0.1690351
```

B.5 Weibull model and AIC comparison

```
               Value Std. Error     z       p
(Intercept)  6.75737    0.43508 15.53 < 2e-16
age         -0.00889    0.00688 -1.29   0.196
sexFemale    0.41616    0.12698  3.28   0.001
ph.ecog     -0.36643    0.09040 -4.05 5.0e-05
wt.loss      0.00613    0.00473  1.30   0.194
Log(scale)  -0.33381    0.06375 -5.24 1.6e-07

Scale= 0.716
Loglik(model)= -1047.5   Loglik(intercept only)= -1062.6
        Chisq= 30.14 on 4 degrees of freedom, p= 4.6e-6
n= 213

                  df      AIC
exponential_model  5 2128.132
weibull_model      6 2106.980
lognormal_model    6 2136.391
```